A Capable, Fully Tensor-Decomposable Vision-Language-Action Policy

Exact rational conversion, protocol-aligned control, and weight-derived causal structure

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Abstract

Can a capable robot policy be built entirely from learned operations that admit direct tensor decomposition? We introduce χ-VLA, a compact vision-language-action policy whose learned map uses bilinear attention, bilinear feed-forward blocks, linear projections, and rational per-instance normalization. The complete deployed map is therefore a rational tensor program. A mechanical audit of a trained checkpoint finds no softmax, standard activation, or ordinary normalization operation, and explicit reconstructions match deployed computation near the float32 precision floor. On all ten LIBERO-Object tasks, a 20.1M-parameter policy trained from scratch reaches $92.8\% \pm 2.1\%$ with a ViT encoder and $93.7\% \pm 0.8\%$ with a convolutional encoder under the published 280-step, 50-trial, three-seed protocol. Averaging three convolutional checkpoints reaches 96.2% at three times the inference cost. These results establish competitive specialist capability, not broad robotic generalization. We then ask what the algebraic restriction provides. We derive an exact weight-based decomposition of softmax-free bilinear attention, including the rational query-key normalization used by the policy. In a fixed-control screen of 36 trained block-head pairs, all 36 favor the weight-derived rank-128 subspace over equal-rank random subspaces. The median fidelity ratio is 2.44, the mean is 3.03, and the maximum is 10.14. At the policy's visual bond, retaining 64 of 384 causally ranked dimensions preserves $18/20$ closed-loop successes, compared with $19/20$ for the full representation and $0/20$ for a random subspace. An expanded 80-trial counterfactual instruction test nevertheless exposes a scene-identity shortcut. The present evidence establishes capability and exact layerwise structure. It does not yet establish negligible cost against a matched conventional twin, semantic diagnosis from weights alone, or selective coefficient-level repair.

1 Introduction

Most robot policies mix linear maps with softmax attention, smooth activations, and input-dependent normalization. These operations are effective, but they prevent the complete learned map from being represented as one explicit polynomial or rational tensor network. Prior work showed that bilinear language and vision models expose useful weight structure [3, 4]. It remained unclear whether the same restriction could support a continuous closed-loop robot policy.

The control setting is demanding for two reasons. First, small action errors compound when the policy observes and corrects its own trajectory. Second, per-instance normalization and multimodal continuous actions do not transfer automatically from the language-model setting. A token model

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can place a discrete sample outside its learned logit map. A continuous policy must preserve precise actions while avoiding an invalid average of distinct valid modes.

We study the narrower but testable question:

> *Can a specialist VLA retain competitive closed-loop capability when every deployed learned operation is polynomial or rational, and does the resulting structure provide a useful weight-based analysis surface?*

The stronger thesis motivating the next experiments is a three-part conjunction. A compact class of specialist manipulation policies should incur no more than a predeclared capability loss relative to matched conventional twins. Its trained weights should expose stable behaviorally causal interaction terms. Direct edits to those terms should then produce predicted and selective changes in closed-loop behavior without retraining. Each component has precedent in isolation. Their conjunction in a competitive visuomotor policy is the intended contribution.

Our contributions are:

1. We construct a complete VLA from tensor-compatible primitives. A rational normalization supplies input-specific scaling without leaving the rational function class.
2. We evaluate two fully rational 20.1M-parameter policies under a protocol aligned with published LIBERO-Object results. The strongest single architecture reaches $93.7\% \pm 0.8\%$ over three seeds, and a three-checkpoint ensemble reaches 96.2%.
3. We mechanically audit the real trained policy and explicitly reconstruct deployed rational and bilinear branches from their weights.
4. We extend exact weight-derived decomposition through individual attention layers with a reduced-Gram calculation that avoids materializing the full high-order tensor.
5. We show a causal closed-loop consequence. A selected 64-dimensional visual subspace preserves almost all tested behavior, while a random equal-width subspace fails. A separate instruction intervention exposes a real scene-identity shortcut and limits the language-grounding claim.

Table 1 separates the evidence already established from the tests still needed for the stronger thesis. This distinction is important because algebraic representability does not by itself imply tractable analysis, and a causally important activation subspace is not yet a coefficient-level editing interface.

Conjunct	Current evidence	Decisive missing test
Near-zero capability cost	Two fully rational 20.1M policies reach 92.8% and 93.7% mean success.	Same-skeleton conventional and partial-conversion twins under paired budgets.
Weight-only diagnosis	Exact layerwise attention decomposition consistently beats fixed random subspaces.	Recover a concrete behavioral condition from weights before viewing its test rollouts.
Predictive surgery	A selected activation subspace causally preserves behavior.	Edit specified coefficients, predict the direction, and measure target effect and collateral damage without retraining.

Table 1: **Status of the three-part thesis.** Strong capability and exact layerwise structure are established. The matched cost and coefficient-surgery claims remain open.

This paper does not claim current state of the art. It also does not claim that comparable mechanisms are impossible to obtain from conventional policies. The empirical question is whether the algebraic policy makes them more direct, stable, and selectively editable than matched post-hoc methods.

2 A rational tensor robot policy

2.1 Tensor-compatible primitives

Homogeneous coordinate. We write $\bar{x} = [1; x]$. The constant coordinate allows a multiplicative layer to represent bias, linear, and quadratic terms in one contraction.

Bilinear feed-forward block. For $L, R \in \mathbb{R}^{r\times(d+1)}$ and $D \in \mathbb{R}^{d\times r}$,

$$\mathrm{BFFN}(x) = D\left[(L\bar{x}) \odot (R\bar{x})\right], \qquad y_o = \sum_{i,j}\left(\sum_{\rho} D_{o\rho} L_{\rho i} R_{\rho j}\right)\bar{x}_i \bar{x}_j. \tag{1}$$

The three matrices are a CP factorization of the order-three coefficient tensor.

Softmax-free bilinear attention. Each head forms two query-key systems and one value stream:

$$Y = C\left[M \odot (Q_1 K_1^\top) \odot (Q_2 K_2^\top)/s\right] V. \tag{2}$$

Here M is a fixed causal mask, C is fixed row scaling, and s is fixed. All learned query, key, value, and output maps are linear. Equation 2 is therefore a polynomial in its input activations.

Rational per-instance normalization. RMS normalization contains an input-dependent square root. We instead deploy a fixed rational approximation $r(z) = P(z)/Q(z)$ to $1/\sqrt{z+\epsilon}$:

$$\mathrm{RNorm}(x) = x\, r\left(\frac{1}{d}\sum_{j=1}^{d} x_j^2\right). \tag{3}$$

The deployed function is exactly rational. Projective coordinates carry each activation as a pair (N, D) representing N/D. Linear, bilinear, residual, and normalization operations update the numerator and denominator without introducing a non-rational learned operation.

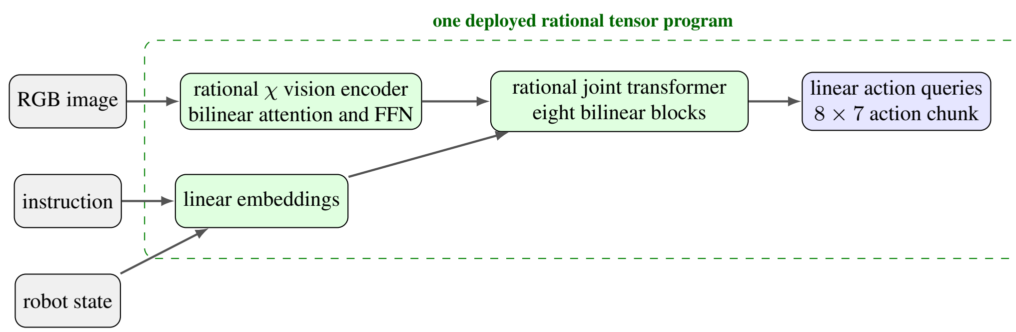

Figure 1: **Deployed χ-VLA map.** The strongest checkpoints use one vision stream and a linear continuous-action head. Every learned component inside the dashed boundary is linear, bilinear, or rational.

2.2 Architecture and training

The policy follows the high-level VLA sequence of visual encoding, language and state embedding, joint processing, and action decoding [1]. It uses a 64×64 agent-view image, an eight-dimensional robot state, a 26-token task vocabulary, and eight action-query tokens. The joint backbone has width 384, eight blocks, and twelve heads. The action head maps each action-query state directly to seven continuous action dimensions. The complete model has $20{,}137{,}352$ learned parameters.

Training uses $66{,}984$ demonstration frames, $40{,}000$ AdamW updates, batch size 256, cosine decay, bfloat16 arithmetic, and an exponential moving average. Closed-loop evaluation uses a $180°$ camera rotation to match the demonstration convention, canonical LIBERO initial states, and fail-loud checks that verify every requested state was loaded.

2.3 Mechanical certificate on the trained checkpoint

Architecture definitions can differ from deployed code. We therefore audit the real $189/200$ checkpoint at runtime and reconstruct representative operations from saved weights.

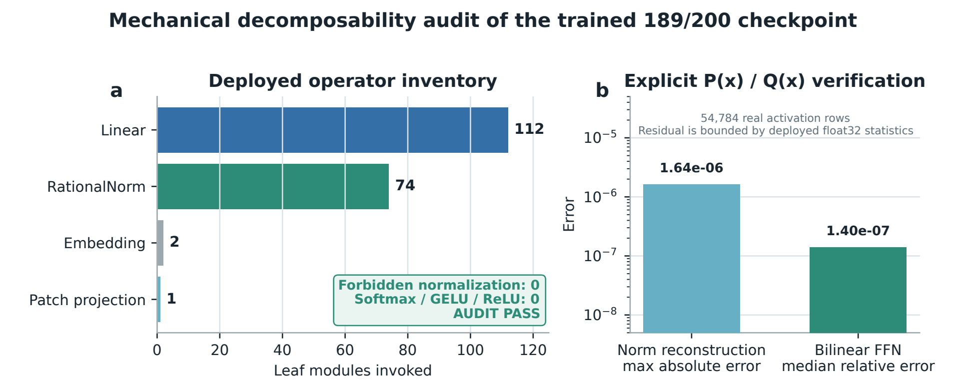

Figure 2: **Mechanical audit of the trained rational checkpoint.** The runtime path invokes 112 linear maps, 74 rational normalizers, two embeddings, and one linear patch projection. No softmax, standard activation, or ordinary normalization is present. Reconstruction errors are measured over $54{,}784$ real activation rows.

The normalization reconstruction has maximum absolute error 1.64×10^{-6}. A bilinear FFN reconstruction has median relative error 1.40×10^{-7}. These residuals are bounded by the deployed module's float32 statistic. The audit certifies operator membership and local reconstruction. It does not materialize one compact polynomial for the full eight-layer transformer.

3 Protocol-aligned closed-loop capability

3.1 Evaluation protocol

We evaluate all ten LIBERO-Object tasks with:

- 50 canonical trials per task
- a 280-step cap
- three independently trained seeds per architecture
- 500 rollouts per seed and $1{,}500$ rollouts per architecture

Published OpenVLA and Diffusion Policy results are protocol-aligned references. They were not rerun in our codebase, so differences may still include implementation and training choices.

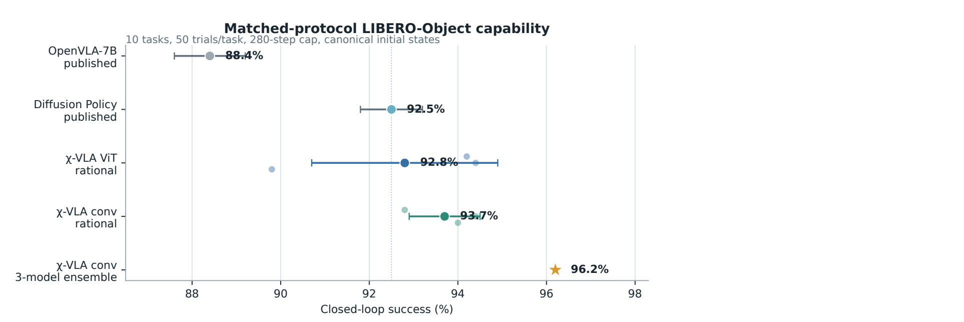

Figure 3: **Protocol-aligned LIBERO-Object capability.** Small dots are χ-VLA training seeds. The ensemble star averages three convolutional checkpoints and uses three forward passes per action chunk. External references are reported published values.

3.2 Results

Method	Parameters	Training seeds	Success
χ-VLA, rational ViT	20.1M	3	$92.8\% \pm 2.1\%$
χ-VLA, rational conv	20.1M	3	$\mathbf{93.7\%} \pm \mathbf{0.8\%}$
χ-VLA, conv ensemble	3×20.1M	$3 \to 1$	$\mathbf{96.2\%}$
OpenVLA-7B [1]	7B	3	$88.4\% \pm 0.8\%$
Diffusion Policy [1]	not matched	3	$92.5\% \pm 0.7\%$

Table 2: **Closed-loop success under aligned evaluation settings.** The two external rows are historical references rather than current state-of-the-art claims.

The ViT seeds score 89.8%, 94.4%, and 94.2%. The convolutional seeds score 94.0%, 94.4%, and 92.8%. The convolutional model is modestly higher and more consistent in this three-seed comparison.

Elementwise averaging of the three convolutional action predictions reaches 96.2%. This is 2.3 points above the mean of its members and 1.2 points above the strongest member. The gain is largest on tomato sauce, butter, and ketchup. The ensemble requires three complete forward passes and is not one 20.1M-parameter policy.

3.3 What this result does and does not establish

The result shows that the rational architecture supports high closed-loop specialist capability. It does not isolate the cost of decomposability. A decisive cost experiment must train a same-skeleton conventional transformer and staged partial-conversion controls using identical data, updates, seeds, and evaluation states. We treat that experiment as required before restoring the stronger title claim that convertibility is nearly free.

4 Exact weight-derived structure through attention

4.1 Exact layerwise construction

Equation 2 contains two query-key products and one value stream. Expanding the linear maps gives a high-order coefficient object determined entirely by the weights, mask, and fixed scaling. On a tiny four-token, width-four, one-head layer, direct evaluation and an independent explicit polynomial agree to 2.00×10^{-15}. The internal core identity agrees to 1.24×10^{-14}.

The deployed attention also normalizes each query and key with Equation 3. Each normalization multiplies all channels of one token by the same scalar rational factor. Those factors can be pulled outside the bilinear dot products. The polynomial attention core is unchanged, while explicit numerator and denominator factors carry the normalization. The rational extension agrees to 3.61×10^{-16}. Agreement with the real trained module ranges from 3.10×10^{-7} to 1.03×10^{-5} because the deployed implementation intentionally casts the statistic to float32.

Naively materializing the real width-384 coefficient tensor would require approximately 3.2×10^{15} values. We instead contract the CP factors into an $O(d^2)$ reduced Gram. At toy multi-head scale, the reduced calculation matches brute-force materialization to 2.13×10^{-14}.

4.2 Trained policy result

We apply the exact weight-derived calculation to all twelve heads in early, middle-late, and final backbone blocks 0, 6, and 7. Random projector banks are sampled once, then fixed and reused for all evaluation examples. Real cached activations are used only to evaluate how faithfully the selected weight-derived subspaces preserve the layer's computation.

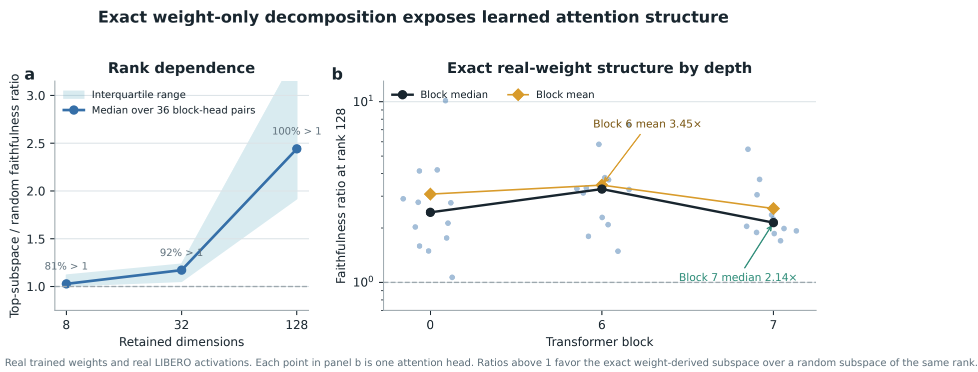

Figure 4: **Exact weight-derived attention structure.** Ratios above one favor the exact weight-derived subspace over a random equal-rank subspace. Each point in the right panel is one trained attention head.

At ranks 8, 32, and 128, respectively, 80.6%, 91.7%, and 100% of the 36 block-head pairs favor the exact subspace. Their median ratios are 1.028, 1.171, and 2.441. At rank 128, the mean ratio is 3.033 and the strongest head reaches 10.135. The rank-128 block means are 3.08, 3.45, and 2.56 for blocks 0, 6, and 7.

This extends exact weight-only construction through individual attention layers. It is not an exact end-to-end decomposition of the complete policy. Composition across all eight blocks still faces rapid degree and representation growth.

5 Causal policy structure and a language shortcut

We next test whether the identified structure affects closed-loop behavior and whether high task success reflects robust use of the language input.

5.1 Causal visual subspace

At the visual bond before the joint backbone, we rank directions separately for translation, rotation, and gripper actions. We compare the selected subspace with random equal-width subspaces in offline action reconstruction and closed-loop intervention. This ranking uses held-out activations and downstream sensitivity. It is causal, but not a weight-only discovery result.

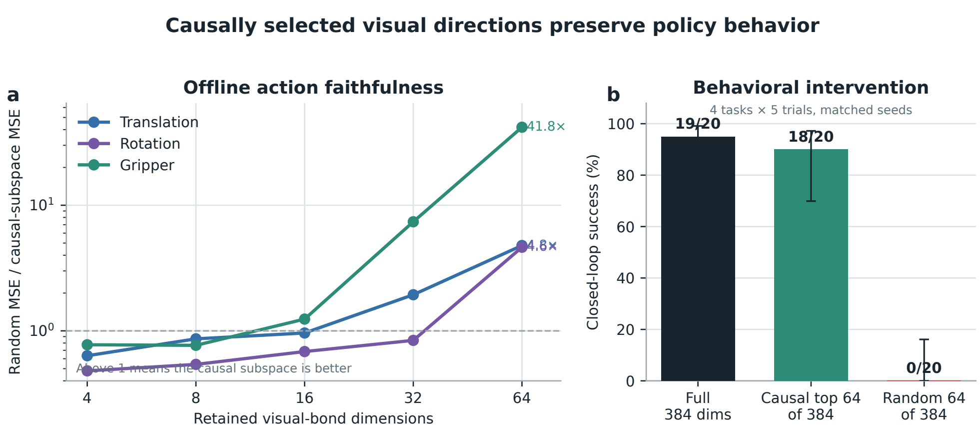

Figure 5: **Causally selected visual directions preserve policy behavior.** The left panel reports random MSE divided by selected-subspace MSE. The right panel intervenes on the real policy over four tasks and five paired trials per task.

With 64 of 384 dimensions retained, the selected subspace reduces action reconstruction error relative to random by $4.8\times$ for translation, $4.6\times$ for rotation, and $41.8\times$ for the gripper. In closed loop, the full representation succeeds in $19/20$ trials, the selected subspace in $18/20$, and a random subspace in $0/20$.

This result establishes a causal low-dimensional substrate, but it is still limited to one checkpoint and twenty rollouts. The current ranking at this full-policy bond uses held-out examples and downstream sensitivity. The exact layerwise attention method in Section 4 must still be connected to the same full-suite closed-loop intervention.

5.2 Counterfactual instruction test

High success does not guarantee robust language use. We hold the image and robot state fixed and replace the instruction with one naming another visible object. Across 80 canonical one-step trials, the predicted reach direction shifts toward the newly named object in only 15.0%. Mean directional cosine to the named object falls from 0.525 under the true instruction to 0.299 under the counterfactual instruction, while cosine to the original target remains 0.464. In an earlier 40-step closed-loop screen, 33.3% finish closer to the new object and 55.6% move measurably toward it.

The likely shortcut is structural in the dataset. Each scene template is paired with an almost fixed target, so scene identity can substitute for object-name grounding. Decorrelated scripted demonstrations improve the counterfactual metric, but all three fine-tuning attempts severely reduce ordinary closed-loop success. The best repaired dataset raises capability only to 21.5%, far below the healthy range.

This negative result narrows the VLA claim. The current model is a capable language-conditioned specialist, but it is not yet a robust instruction-grounded policy. LIBERO-CF independently documents the same broad failure class in existing VLAs [10]. Our prospective novelty is therefore not the shortcut alone. It is the harder test of whether exact weight-derived terms can localize and selectively repair that shortcut.

6 Decisive remaining tests

1. Controlled conversion cost. Train a same-skeleton conventional model with softmax attention, a standard gated FFN, and RMSNorm. Add partial-conversion controls. Use paired seeds and predeclare a three-point noninferiority margin. Match observations, token layout, action decoder, demonstrations, augmentation, optimizer, updates, evaluation states, and hyperparameter-search budget. Report parameters, FLOPs, throughput, memory, and latency. This is the missing experiment for any claim that convertibility is nearly free.

2. Direct coefficient surgery. Our first preregistered four-task screen directly edits query, key, and value columns selected by exact attention Grams. Retention gives a favorable 10-point gap over random controls but misses the 15-point gate, while removal has the wrong sign. This is a no-go for the current edit mapping, as detailed in Appendix A.5. The next test must resolve redundant paths or gauge sensitivity, compare post-hoc controls across three checkpoints, and report targeted effect, ordinary success, edit norm, and collateral damage.

3. Concrete diagnosis and selective repair. Develop two capstones. Gripper timing is the lower-risk target because phase transitions and premature closure have explicit simulator variables. Scene identity versus instruction binding is the higher-upside target because the current policy demonstrably fails the counterfactual test. In both cases, discover the algebraic term from weights, attach a limited semantic label, freeze the candidate, predict the behavioral consequence, and test on blind paired rollouts. A compelling repair should improve the targeted failure by 10 to 20 points, lose no more than 2 to 3 points of ordinary success, and preserve its direction across three seeds.

4. Broader capability and grounding. Evaluate the conventional and rational finalists on LIBERO-Goal and LIBERO-Long, followed by a counterfactual grounding benchmark such as LIBERO-CF [10]. Object-only success is insufficient because standard LIBERO can reward scene-to-action memorization [9]. Test the stability of recovered mechanisms under equivalent reparameterizations and across trained seeds.

5. RL only after a paired gate. The repaired-reward AWR pilot improves 80.7% to 82.7%, but it has one seed and fifteen trials per task. A justified follow-up needs three paired seeds, a matched-compute behavior-cloning continuation control, full protocol evaluation, and regression gates. Generic PPO on saturated LIBERO-Object is lower priority than the four tests above.

7 Related work

VLA capability. OpenVLA established an open pretrained VLA and standardized LIBERO comparisons [1]. OpenVLA-OFT showed that continuous action chunks and simple regression can substantially improve adaptation and control efficiency [2]. These systems use large pretrained backbones. χ-VLA instead studies a compact from-scratch specialist under an algebraic restriction.

Tensor and polynomial networks. Polynomial networks and tensor networks use explicit multilinear structure for representation and learning [12–14]. Bilinear MLP work showed that weight eigendecomposition can expose low-rank features and circuits [3]. χ-nets introduced ODT for compositional bilinear networks [4]. We extend the setting to a continuous closed-loop policy and give an exact reduced construction for individual softmax-free attention layers.

Causal mechanisms and policy analysis. Bilinear IMPALA policies have combined competitive ProcGen control with weight-derived eigenfilters, followed by activation probing and control [5]. This is the closest architectural precedent for weight-based policy analysis. Recent work steers pretrained VLAs through causally identified activation directions [6]. Sparse feature circuits identify and edit causal feature graphs in conventional language models [7], while nonlinear causal reductions recover behavioral patterns in reinforcement-learning policies from rollout perturbations [8]. These results make a universal separation from conventional policies untenable. Our intended distinction is empirical: exact weight-derived discovery and direct coefficient edits in a competitive continuous vision-language policy, evaluated for stability and collateral damage against matched post-hoc baselines.

VLA robustness. Recent evaluations show that standard LIBERO success can coexist with weak robustness and language use [9–11]. Our instruction intervention reaches the same conclusion from an independently trained compact specialist and motivates a decomposition-guided diagnosis.

8 Limitations and conclusion

χ-VLA demonstrates that a complete rational tensor policy can reach strong specialist closed-loop capability. The real checkpoint passes a mechanical operator audit. Exact weight-derived analysis now crosses individual attention layers, and a compact causal visual subspace preserves almost all tested behavior.

Five limits determine the next version of the paper. First, no same-skeleton conventional twin has yet measured the exact capability and compute cost of conversion. Second, the benchmark is one narrow suite and the policy fails a counterfactual instruction test. Third, exact decomposition is layerwise rather than end to end. Fourth, the closed-loop causal intervention uses one checkpoint and twenty trials. Fifth, activation projection succeeds but the first direct coefficient-edit screen fails its selectivity gate, so no behaviorally validated weight edit has yet been established.

The defensible conclusion is therefore narrow. Polynomial and rational restrictions do not prevent a compact policy from reaching competitive LIBERO-Object capability, and they provide a precise weight-derived analysis surface. Establishing negligible conversion cost, broad language-grounded control, semantic diagnosis from weights, and a decomposition-enabled repair requires the controlled experiments in Section 6. If all three conjuncts in Table 1 hold on the same trained policies, the result is stronger than any individual capability, decomposition, or intervention claim.

Reproducibility. Training and evaluation use fixed task lists, explicit camera and action conventions, canonical-state manifests, checkpointed moving-average weights, and per-run result files. The final artifact will release code, checkpoint hashes, exact attention certificates, and per-episode evaluation logs. Preflight checks fail if canonical initial states cannot be loaded.

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A Supporting results and decisions

A.1 Original 94.5% capability result

Before the final protocol-aligned benchmark, one rational ViT checkpoint succeeded in $189/200$ trials under a 600-step horizon and 20 trials per task. This result established that the rational implementation could support all ten tasks. It should not be compared directly with the protocol-aligned external references because it uses one training seed, fewer trials, and a longer horizon.

A.2 Per-task ensemble behavior

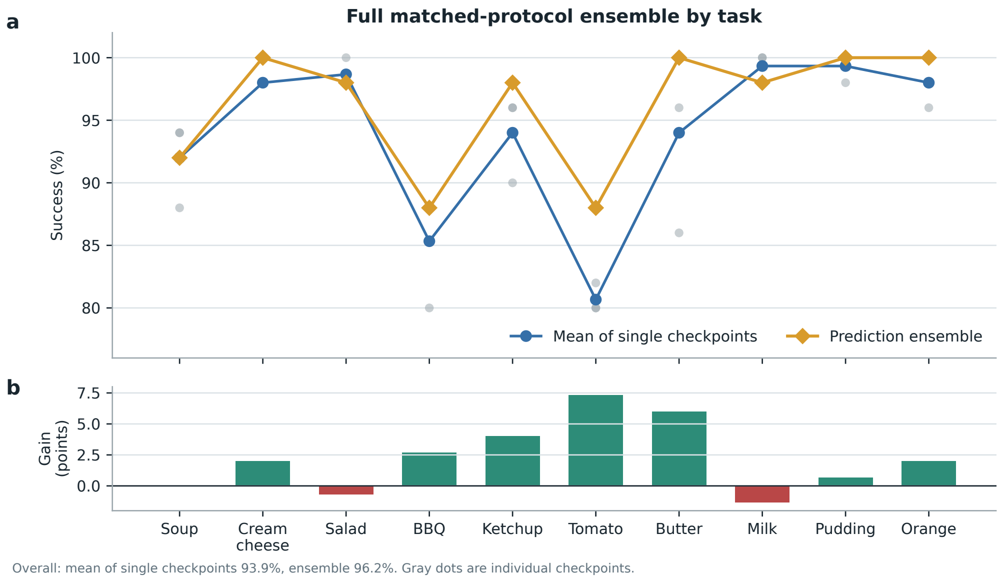

Figure 6: **Per-task effect of prediction averaging.** Gray points are the three convolutional checkpoints, evaluated with 50 canonical trials per task. Curves compare their taskwise mean with elementwise action-prediction averaging. The ensemble raises overall success from a 93.9% member mean to 96.2%, with its largest gains on tomato sauce, butter, and ketchup. It requires three forward passes and is not a single 20.1M-parameter policy.

A.3 Product-routing multimodal head

For an action-query state h, the product-routing head represents

$$a(h,b) = c_0(h) + \sum_{g=1}^{G} (2b_g - 1) c_g(h), \qquad b \in \{0,1\}^G. \tag{4}$$

The center, offsets, and gates are bilinear CP modules. A final external sign choice selects one of 2^G modes. The head folds to numerical precision, but its five-seed closed-loop mean is $44\% \pm 30\%$ and its range overlaps the linear baseline. The strongest rational policy therefore uses the linear head. Product routing remains a construction result until it wins a controlled genuinely multimodal task.

A.4 Development interventions

Focused DAgger on BBQ and tomato improves combined success from 82.5% to 96.0% using seven corrective episodes and $1{,}131$ frames. Applying the same update thinly across all ten tasks reduces overall success from 92.0% to 72.0%. Three original AWR runs decline by 8.0, 4.6, and 2.6 points. A reward audit shows that accumulated step cost makes every successful trajectory have negative return. After removing that term and repairing the terminal boundary, one pilot improves by two points. These are development diagnostics, not validated paper contributions.

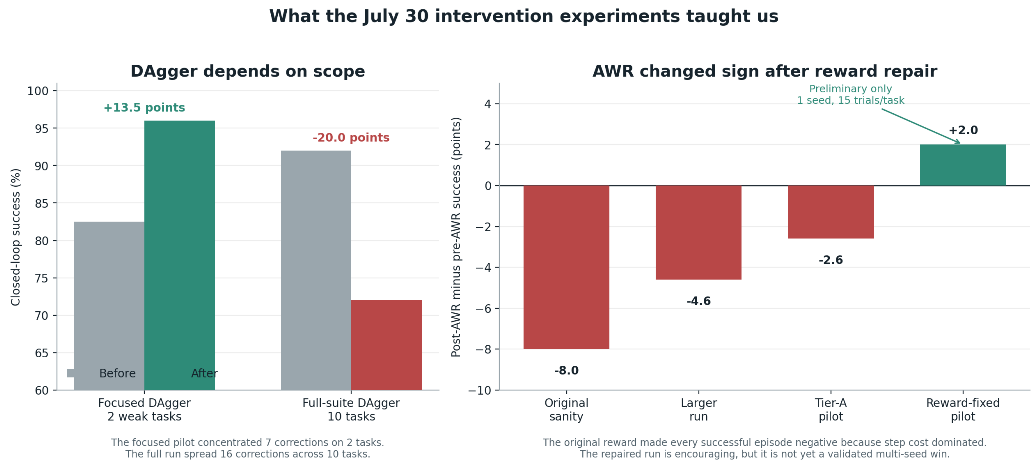

Figure 7: **Development interventions reveal scope and reward sensitivity.** Focused DAgger helps two weak tasks, while spreading a small correction budget across ten tasks causes regression. Three AWR variants also regress before a reward audit and repair. The positive reward-repaired AWR result is a one-seed pilot with 15 trials per task, so it is not evidence of a validated improvement.

A.5 Expanded attention screen and first direct edit

The corrected fixed-control attention analysis covers all eight blocks and twelve heads. At rank 128, $83/96$ block-head pairs favor the exact weight-derived subspace. The median ratio is 1.94, the mean is 2.58, and blocks 2 and 3 provide useful weak or negative controls. At ranks 8 and 32, only $54/96$ and $60/96$ pairs favor the exact subspace. The effect is therefore depth-dependent and strongest at the wider tested rank.

The first direct weight-edit screen uses four tasks and five paired canonical episodes per task. Exact-subspace retention preserves $18/20$ successes compared with $16/20$ for both random controls. This 10-point advantage misses the preregistered 15-point gate. Exact removal also preserves $18/20$, compared with $17/20$ for both random controls, which is the wrong direction for the necessity prediction. The screen is a no-go for the current mapping from Gram eigenspaces to query, key, and value column edits.

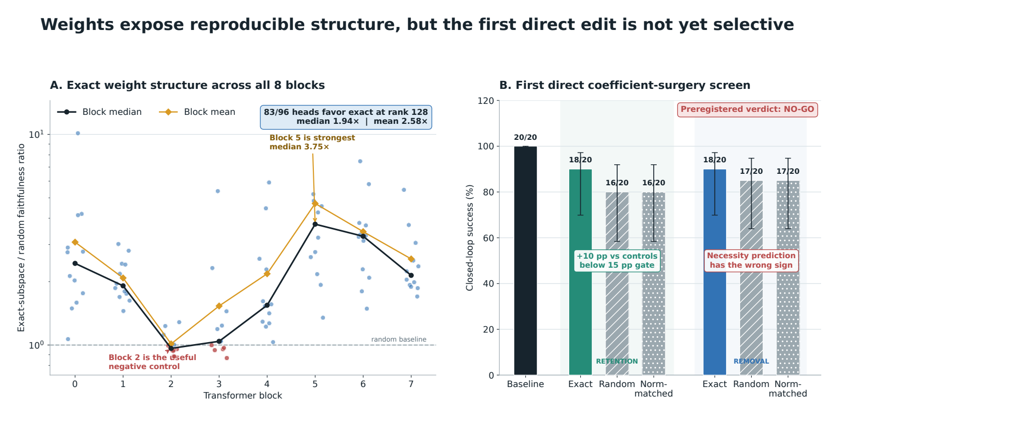

Figure 8: **Expanded weight-structure and direct-surgery results.** Left: all 96 block-head pairs at rank 128, with fixed random-control banks. Exact directions come from weights, while cached activations only score layer faithfulness. Right: the first direct query-key-value column-edit screen over four tasks and five paired trials per task. Error bars are 95% Wilson intervals. The preregistered verdict is no-go because retention misses the effect-size gate and removal has the wrong sign.

A.6 Numerical scope of rational conversion

The deployed Padé computation is exactly rational even though it approximates RMS normalization. Final certification must report its polynomial degrees, coefficients, denominator margin, observed activation range, tail error, action drift, and runtime overhead. Operator membership alone does not establish practical numerical safety under distribution shift.